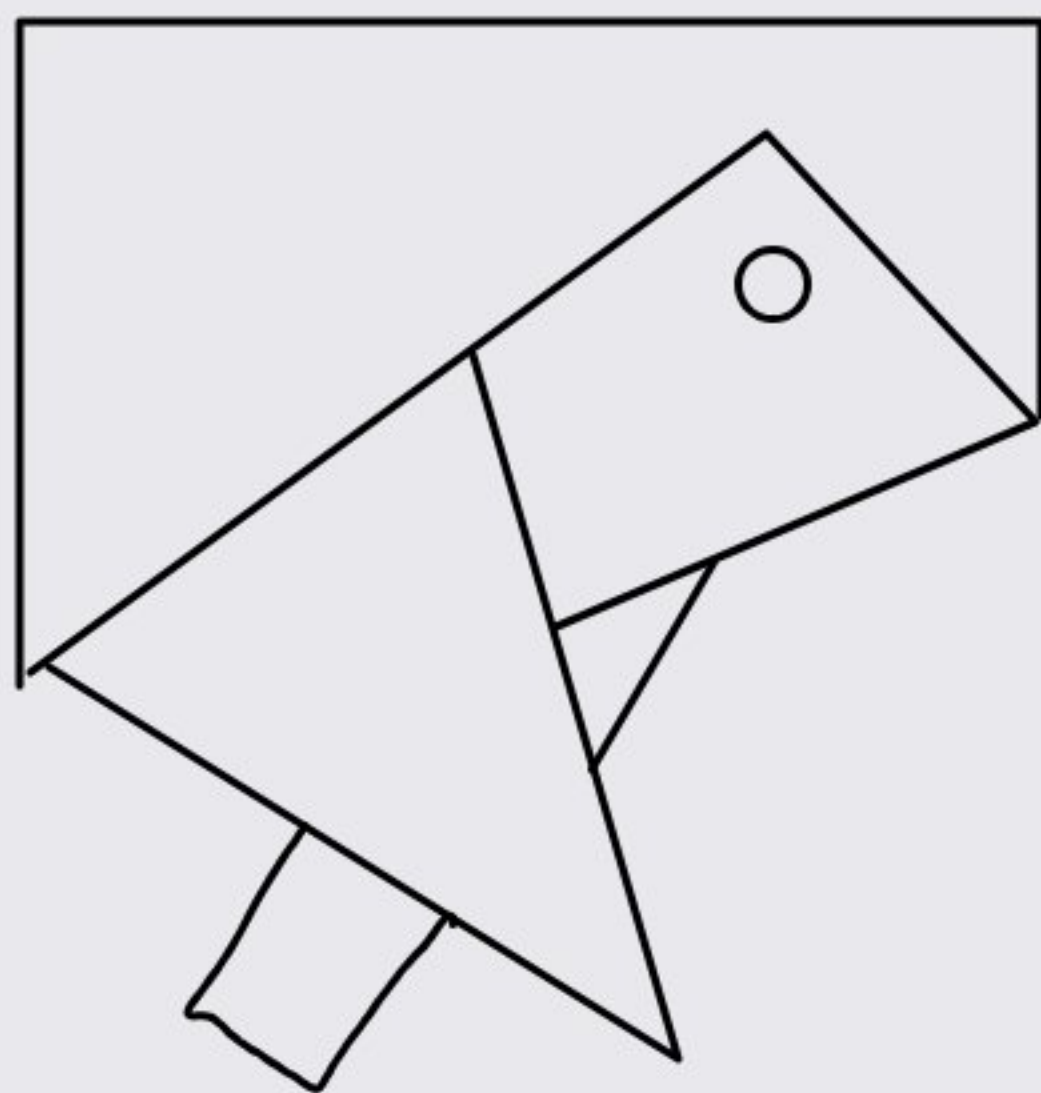
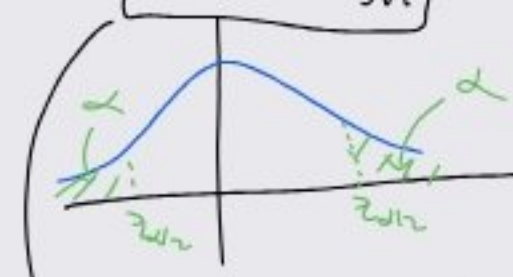




Intervalo de confianza para μ :

$$\hat{\theta} = \left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] \leq \mu \leq \left[\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] = \hat{\theta}_n$$



Con un $(1-\alpha) \cdot 100\%$ de confianza, la media poblacional μ se encuentra en el intervalo $\left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$.

Límites de confianza unilaterales:

La μ poblacional esté contenida en un valor:

(1) mayor o igual que $\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$, $(1-\alpha) \cdot 100\%$

(2) menor o igual que $\bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$.



Ejemplo. En un estudio de pruebas psicológicas, se seleccionan al azar 25 sujetos y se mide su tiempo de reacción, en segundos, ante una prueba en particular. La experiencia pasada sugiere que la varianza de tiempo de reacción a dichas pruebas es de $4s^2$ y que el tiempo de reacción es aproximadamente normal. El tiempo promedio para los sujetos fue de 6.2s. Dé un límite superior del 95% de confianza para el tiempo medio de reacción.

Sol.

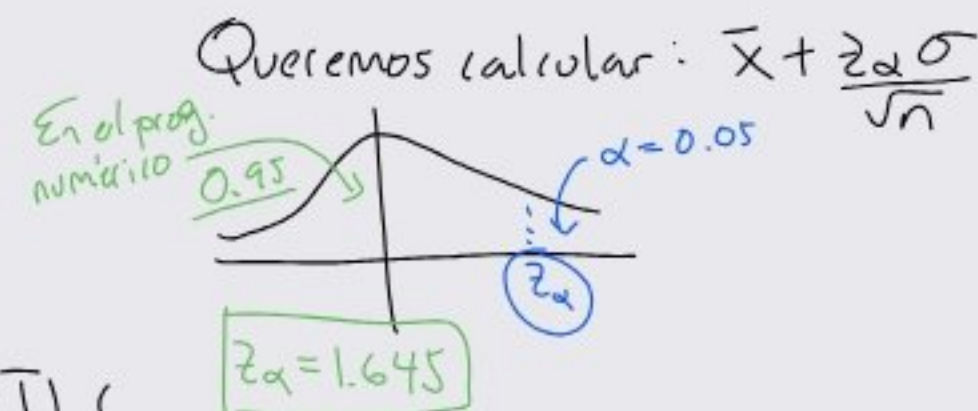
$$n = 25$$

$$\sigma^2 = 4$$

$$\bar{x} = 6.2$$

$$\alpha = 0.05$$

Pob. normal $\rightarrow$ TL



El límite superior es: $\bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}} = 6.2 + \frac{(1.645) \cdot 2}{5} =$

l_s

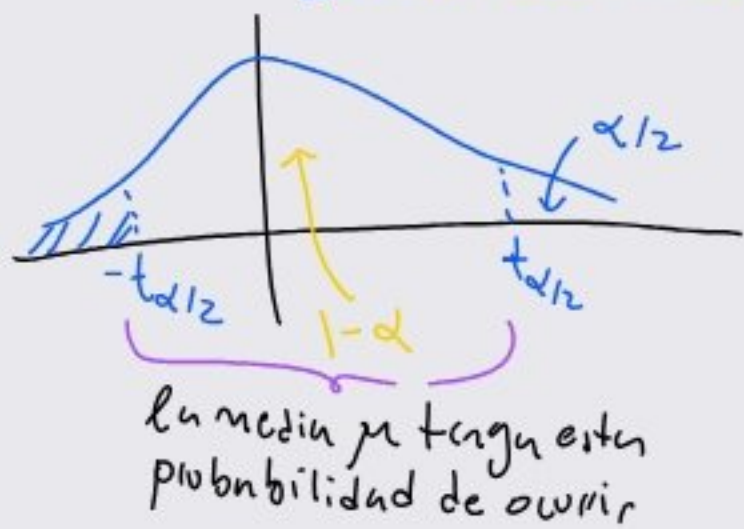
R. Con el 95% de confianza, la media poblacional μ es menor a l_s .

Límites de confianza con σ -desconocida:

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

desv. est. muestral

Límite de confianza es $(1-\alpha) \cdot 100\%$



$$P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1-\alpha$$

$$P(-t_{\alpha/2} < \frac{\bar{X} - \mu}{s/\sqrt{n}} < t_{\alpha/2}) = 1-\alpha$$

$$-t_{\alpha/2} < \frac{\bar{X} - \mu}{s/\sqrt{n}}, \text{ despejar } \mu$$

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} < t_{\alpha/2}$$

$$-t_{\alpha/2}(s/\sqrt{n}) < \bar{X} - \mu \quad \bar{X} - \mu < t_{\alpha/2}(s/\sqrt{n})$$

$$\mu < t_{\alpha/2}(s/\sqrt{n}) + \bar{X}$$

$$\mu < \bar{X} + t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$\mu > \bar{X} - t_{\alpha/2}(s/\sqrt{n})$$

$$\mu > \bar{X} - t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$\bar{X} - t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Intervalo de confianza del $(1-\alpha) \cdot 100\%$ para μ con σ -desconocida.

$$\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Error estándar: relación entre estimaciones puntuales y de intervalo.

$\bar{X}$: estimación puntual de μ

$$\left[\bar{X} - e.e., \bar{X}, \bar{X} + e.e. \right]$$

C.E.: error estándar.

$$C.E.(\bar{X}) = \frac{\sigma}{\sqrt{n}} \rightarrow \bar{X} \pm z_{\alpha/2} \cdot e.e.(\bar{X})$$

$$\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

estimación por intervalo de μ con σ -conocida.

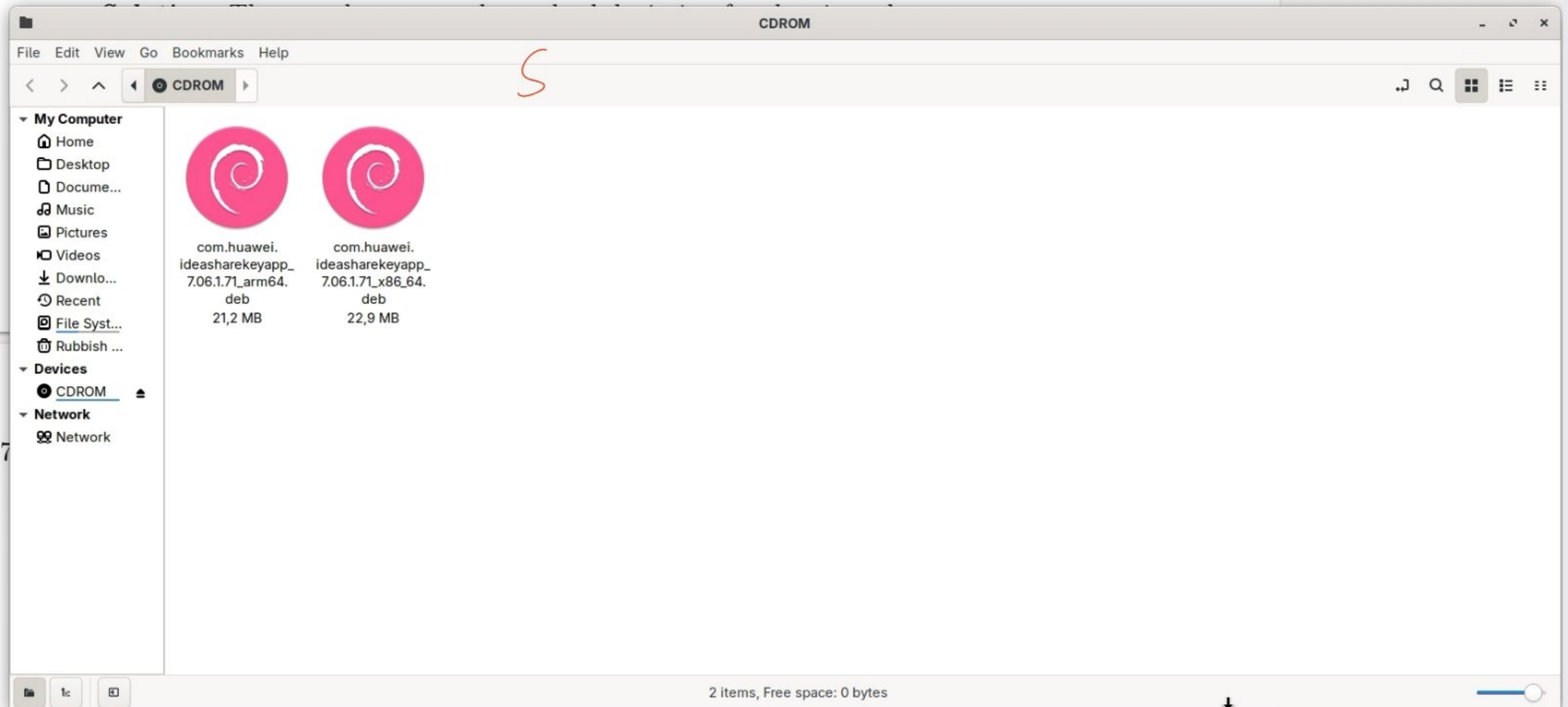
$$\bar{X} - t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

estimación por intervalo de μ con σ -desconocida.

$$C.E.(\bar{X}) = \frac{s}{\sqrt{n}} \rightarrow \bar{X} \pm t_{\alpha/2} \cdot e.e.(\bar{X})$$

They are the upper and lower $100(1 - \alpha)/2\%$ bounds, respectively. Here t_α is the t -value having an area of α to the right.

Example 9.5: The contents of seven similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% confidence interval for the mean contents of all such containers, assuming an approximately normal distribution.



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Example 9.5: The contents of seven similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% confidence interval for the mean contents of all such containers, assuming an approximately normal distribution.

Sol:-

$n = 7; v = 6$
 $\alpha = 0.05$
 $\alpha/2 = 0.025$

$x_1 = 9.8$
 $x_2 = 10.2$
 $x_3 = 10.4$
 $x_4 = 9.8$
 $x_5 = 10.0$
 $x_6 = 10.2$
 $x_7 = 9.6$

$\bar{x} = \sum_{i=1}^7 x_i = 10; s = ?$

$\bar{x} - t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$

$1 - 0.025 = 0.975$
 $t_{0.025}$
 $f(t; 6) dt = 0.975$

$-t_{0.025}$ $t_{0.025}$

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